Name: _____ Student ID: _____

Suppose you roll a six-sided die with a probability distribution given below. Let $X$ be the number of spots rolled.

| $x$ | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{10}$ | $\frac{3}{10}$ | $\frac{2}{10}$ | $\frac{1}{10}$ | $\frac{2}{10}$ | $\frac{1}{10}$ |

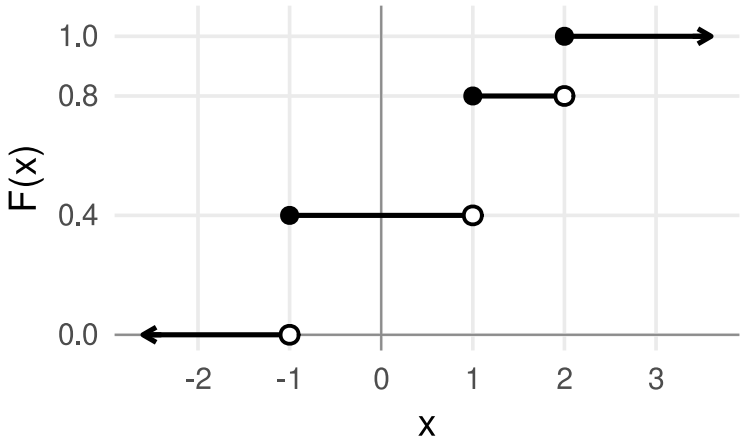
1. Calculate $E(X)$

2. Calculate $Var\left(\frac{1}{2}X + 300\right)$.

3. Let $X$ be a discrete uniform random variable on $\{-4, -1, 0, 1, 2\}$. Let $Y = X^2$. Find $E(Y)$.

4. Let $X$ be a random variable such that $E(X) = 4$. Let $Y = 2 + 3X$. Find $E(Y)$.

5. Given the cumulative distribution function $F(x)$ of the random variable $X$ below, compute the expected value and variance of $X$ (you will need to find the pmf of $X$ first).



An American roulette wheel has 38 slots: 18 red (`"R"`), 18 black (`"B"`) and 2 green (`"G"`).

6. Let $S$ be the number of times in 10 spins that you land on a **red** space. Calculate $E(S)$, $Var(S)$ and $SD(S)$.

*Hint:* You can simulate a random variable by drawing from a box of tickets. Choose the tickets so that each value appears in the right proportion, use `rep()` to build the box, and draw with replacement. For example, a coin that lands heads with probability 0.7 can be represented by a box of 10 tickets, 7 marked `"H"` and 3 marked `"T"`:

```
box <- c(rep("H", 7), rep("T", 3))
```

and `sample(box, size = 5, replace = TRUE)` simulates 5 tosses of this coin.

Now you can use this approach to do the problems below.

7. Write R code which will allow you to simulate 10 spins of an American roulette wheel (where your possible outcome is either *red* `"R"` or *not red* `"NR"`), and count the number of spins that result in landing on a red space.

8. Write R code which will allow you to simulate 100 of these experiments. Your output should be a vector with 100 observations, where the first element is the number of red spaces landed on in experiment one, and so on.

9. Write code to calculate the mean, variance and standard deviation of the vector you created in the last step. Then, compare them to the theoretical quantities you calculated at the start of this series of problems in one to two sentences.